

Enhancing Virtual Walking in Lying Position: Upright Perception by Changing Self-Avatar's Posture

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ABSTRACT

We aimed to decrease visual-proprioceptive conflict in experiencing virtual walking with the lying posture. An optic flow during an avatar's standing up was presented before virtual walking that was induced by the radial optic flow and foot vibrations. The walking sensation and telepresence slightly increased by the standing-up optic flow, but it did not reach statistical significance. Participants felt as the posture was more matched to the walking avatar with the standing-up optic flow compared to the no-animation condition. These results highlight the potential for posture-informed VR design to improve user experiences in a situation with visual-proprioceptive conflict.

Keywords: walking, lying position, posture perception, sensory integration

Index Terms: • Human-centered computing~Human computer interaction (HCI)~Interaction paradigms~Virtual reality

1 INTRODUCTION

A virtual sensation of walking can be elicited by presenting optic flow and vibrations to the feet simultaneously, and improved by observing self-body avatar walking from a first-person perspective [1-2]. While natural walking is typically performed while standing, these previous studies employed a seated position to expand user diversity. Studies focusing on the postural conflict between the real and virtual environments have been conducted. Saint-Aubert et al. [3] compared standing, sitting, and Fowler's posture (an inclined upper body position), showing that discrepancies between real body and avatar postures decrease walking sensation and the sense of body ownership. Luo et al. [4] assessed the changes in body ownership and presence showed that a 45-degree upper body incline can cause a mismatch between vision and vestibular perception of gravity, leading to perceptual degradation. Nakamura et al. [5] examined the effects of combining visual and vibration stimuli, showing that a fully lying posture can produce a sense of walking comparable to that in a seated position. Based on these studies, we aimed to complement the perception of posture and reduce sensory discrepancies. Therefore, we hypothesized that visually presenting postural change information through self-body avatar animations in a fully lying posture could create an illusion of getting upright stance and enhance the walking experience.

2 METHODS

Twenty naïve participants with normal eye acuity performed the experiment after provided informed consent. The experiment's methods were approved by the Ethical Committee for Human-Subject Research at the Toyohashi University of Technology.

We used a computer and Unity (2020.3.20f1) to present virtual environment to a head-mounted display (HMD: HTC VIVE Pro Eye, 1440 [width] x 1600 [height] pixels for each eye, refresh rate

90Hz) and foot vibrations to four vibro-transducers (Acoupe Lab Vp408) placed each for the left and right forefeet and heels. To minimize outside noise from the vibro-transducers, participants were given white noise (70 dBA) through noise-canceling headphones.

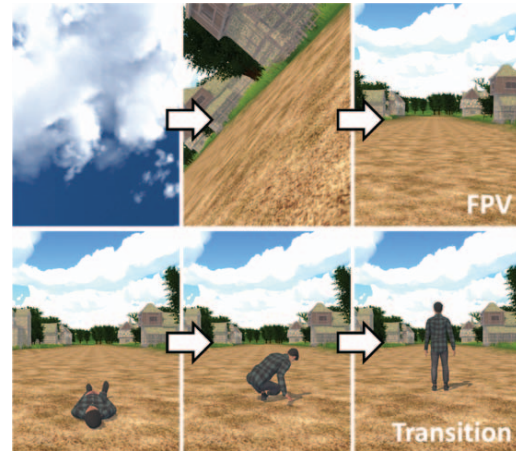


Figure 1: Avatar standing-up Motion. (Top) optic flows presented to participants in the first-person view. (Bottom) avatar's standing-up motion from the third-person view.

A terrain and structure-filled area (250m [width] x 250m [depth]) was created, and a flat walkable area (10m [width] x 120m [depth] x 5m [height]) was set in the center. An area (70m [width] x 140m [depth] x 40m [height]) directly above the walking space was designed to present vertical information by releasing descending particles. These particles were configured using Unity's particle system (1000 particles/s, drop speed 2 m/s). Participants walked across the flat area from one end to the other. Before the avatar began walking, an optic flow during the posture change of the avatar's standing upright was observed for 5 s. Subsequently, an optic flow during walking of the avatar was observed for 65 s.

The avatar walked forward (5.16 km/h, 2.20 steps/s). Participants observed the scene from the avatar's viewpoint (first-person view). The foot vibration stimuli were made by recording the sound of walking in sneakers. A single step was cut out from the recorded data, to make the sound clip of 500 ms. This clip was used for the forefoot and heel with 100 ms gap.

There were three levels of postural animation condition (standing up, 90 deg flip, none) and two levels of vertical fall particles condition (fall, none) (Figure 1). Participants performed 18 trials (3 posture types x 2 particle types x 3 repetitions) in a random order.

Participants were asked to answer a questionnaire after observing a stimulus. The questionnaire consisted of items on walking-related sensations, and items on embodiment.

Walking-related sensations were measured by Visual Analog Scale (VAS); The most left meant no sensation and the most right

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meant the same sensation as in the real environment [1-2]. The data were digitalized to 0-100 after measurements.

W1. I felt as if my whole body was moving forward (self-motion).

W2. I felt as if I was walking forward (walking sensation).

W3. I felt as if my feet were striking the ground (leg action).

W4. I felt as if I was actually there in the scene (telepresence).

The Embodiment Questionnaire section consists of three items selected from previous studies [6]. They are measured using a 7-point Likert scale.

E1. At some point it felt as if my real body was starting to take on the posture or shape of the virtual body that I saw (posture matching).

E2. I felt as if the virtual body was my body (sense of body ownership).

E3. I felt like I could control the virtual body as if it was my own body (sense of agency).

The questions were presented in a random order in each trial.

3 RESULTS

We conducted two-way repeated-measured ANOVAs (3 posture types x 2 particle types). If there was a lack of sphericity with Mendoza's multisample sphericity test, the reported values were adjusted using the Greenhouse-Geisser correction. Shaffer's modified sequentially rejective Bonferroni procedure was applied for post-hoc multiple comparisons.

3.1 Walking-related sensations

For the walking sensation and telepresence scores, we found a marginally significant effect for the posture types [$F(2,38) = 3.18$, $p = .0528$, $\eta_p^2 = 0.143$, $F(1.58, 30.1) = 2.78$, $p = .0887$, $\eta_p^2 = 0.128$], and the standing-up animation increased those scores (Figure 2). However, after conducting the post-hoc test, there was no significant difference. For other items, there was no significant effect.

3.2 Embodiment questionnaire section

For posture matching score, we found a significant interaction between the posture types and the particle types [$F(2, 38) = 3.91$, $p = .0285$, $\eta_p^2 = 0.171$]. The posture type was marginally significant at the no particle condition. Multiple comparisons revealed that participants felt as the posture was more matched with the standing-up animation than without animation when there was no falling particle [$t(19) = 2.79$, $adj.p = .0118$]. For body ownership score, we found a significant main effect for the posture types [$F(2,38) = 3.37$, $p = .0449$, $\eta_p^2 = 0.151$], and the standing up animation increased the sense of body ownership (Figure 3). However, after conducting the post-hoc multiple comparisons, there was no significant difference. For sense of agency score, our analysis revealed no significant effect.

4 DISCUSSION AND CONCLUSION

The observation of optic flow during an avatar's standing-up animation from a lying position slightly increased the walking sensation and the telepresence, but it did not reach the significant effect. This suggests that the contribution of the optic flow of standing-up is weak or limited. Notably, the posture matching assessment showed significantly higher ratings without descending particles, when the optic flow during standing up was observed, compared to no-animation condition. Thus, the visual information of standing up in the first-person perspective could improve the visual-proprioceptive conflict experienced by the lying position participants. Furthermore, the effect of the optic flow on body ownership suggests that the decrement of visual-proprioceptive conflict could contribute to the sense of body ownership.

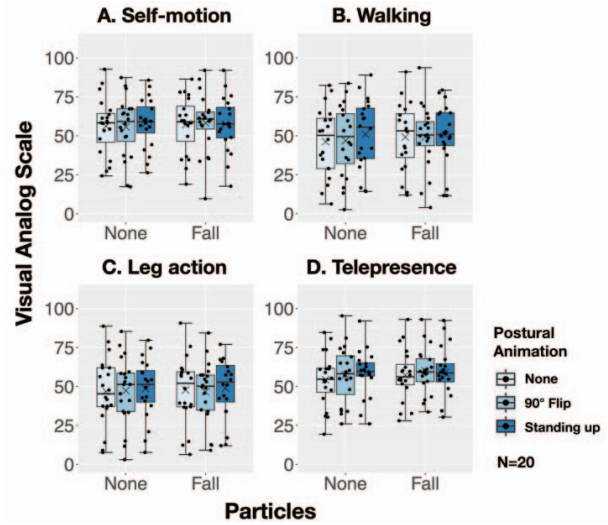


Figure 2: Results of Walking-related sensations.

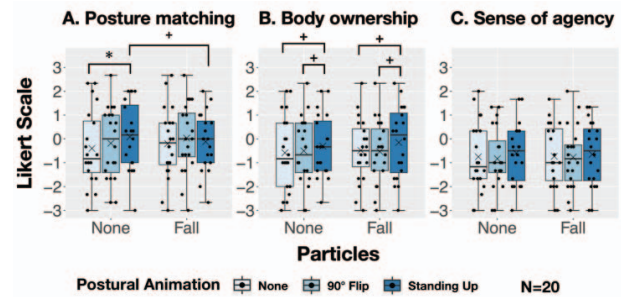


Figure 3: Results of Embodiment questionnaire.

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